

TFE4141 Design of Digital Systems 1

Assignment 4: Datapath design

Mega adder requirements

In this assignment we will design a large adder circuit. This adder circuit must satisfy a set of requirements as described in this section.

Functional requirements

The design takes two 128-bit numbers a and b , add these together and produces a 128-bit result y .

Interface requirements

The design has two interfaces. One interface for sending data in to the design and one interface that is used for sending data out of the circuit.

The input interface is used for sending in the two 128-bit numbers a and b in chunks of 32-bits at a time.

These numbers are added together and produces a 128-bit number y that is sent out of the circuit also in 32-bit chunks.

The 128-bit numbers are chopped up in 32-bit words as illustrated in Table 1.

127..96	95..64	63 ..32	31 ..0
a3	a2	a1	a0
b3	b2	b1	b0
y3	y2	y1	y0

Table 1. Partitioning the 128-bit operands into four 32-bit words.

Both the input and the output interfaces are valid/ready based handshake interfaces.

- data_in/out must have valid data when data_in/out_valid is high.
- Data is transferred on the interface when data_in/out_valid and data_in/out_ready are high both in the same clock cycle.
- If data_in/out_valid is high and the corresponding ready signal is low in a particular clock cycle, then data_in/out_valid must be high also in the following clock cycle and the data_in/out signal must retain its previous value. This situation is illustrated for the data transfer of a_3 and y_3 in Figure 1 and Figure 2.

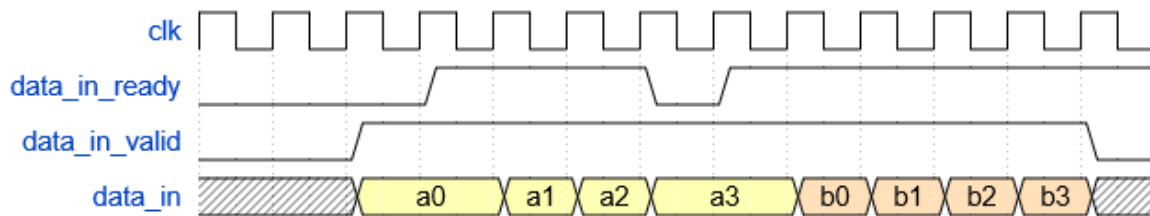


Figure 1. Input interface.

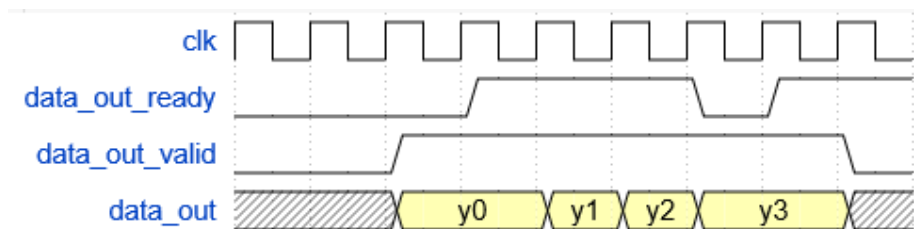


Figure 2. Output interface.

Performance requirements

The design must be able to run at minimum 200MHz.

Task 1: Draw microarchitecture

Examine the code for the mega_adder. Draw the microarchitecture for the data path part of the design. Examine the code of the control part to understand how it controls the data path.

Task 2: Simulate the circuit

Run the simple testbench that has been made for the design. Check that the design produces the correct result.

Task 3: Count the number of flip flops that should be inferred during synthesis

Count the number of flip-flops that will be inferred during synthesis. Do this by counting the number of flip-flops that will be inferred by each process in the design and add up the numbers.

Run synthesis and check that the number of flip flops you counted matches that inferred during synthesis.

Task 4: Synthesize the design and report the max clock frequency.

Synthesize the design by pressing “Run Synthesis”. Constraints have been set up so that the target frequency is 200MHz. The synthesis tool will not be able to meet those constraints. See Appendix A at the end of this assignment to see how to set and find timing constraints.

AFTER HAVING FINISHED TASK 1 THROUGH 4 (NOT BEFORE), YOU MAY LOOK AT THE FIRST PART OF SOLUTION EXAMPLE ON BLACKBOARD.

Task 5: Find the critical path in the design and improve the timing.

Locate the critical path in the design and improve the timing. This will most likely require a change in the microarchitecture. Draw a new version of the microarchitecture for the design that will have a shorter critical path. The solution example for tasks 1 through 4 may give you some inspiration, in case improving the microarchitecture turns out to be hard. The teaching assistants are also an excellent resource for you to use :-)

Task 6: Write RTL code, test and synthesize the new design

Implement the design you suggested in Task 5. Check that it works as expected and that the timing constraints are met.

Appendix A: Reporting timing and editing constraints

The screenshot below shows that the clock period is set up to be 5ns. This corresponds to a clock frequency of 200MHz. When you press the “Report Timing Summary” to the left, the “Design Timing Summary” window will appear.

The “Design Timing Summary” window below shows that we have a negative slack of 1ns. This means that the max frequency is around 166MHz instead of the required 200MHz.

If you want to change the constraints, you can just press “Edit Timing Constraints” and change the period of the clock from 5ns to e.g. 10ns. Hopefully the design will be able to meet those relaxed constraints.

The screenshot displays the Vivado IDE interface. The top toolbar shows the 'Implementation Complete, Failed Timing!' status. The left sidebar contains the 'Project Manager' and 'SYNTHESIS' sections. The 'SYNTHESIS' section is expanded, showing 'Run Synthesis' and 'Open Synthesized Design'. The 'Open Synthesized Design' section is further expanded, showing 'Constraints Wizard', 'Edit Timing Constraints', 'Set Up Debug', 'Report Timing Summary', 'Report Clock Networks', 'Report Clock Interaction', 'Report Methodology', 'Report DRC', 'Report Noise', 'Report Utilization', 'Report Power', and 'Schematic'. The 'Report Timing Summary' option is selected. The main window shows the 'Timing Constraints' dialog box, which is open to the 'Create Clock' tab. The 'Create Clock' tab shows a table with columns: Position, Clock Name, Period (ns), Rise At (ns), Fall At (ns), Add Clock, Source Objects, and Source File. The table contains one row: 1, clk, 5.000, 0.000, 2.500, ☐, [get_ports clk], mega_add. Below the table is the 'All Constraints' section, which shows a table with columns: Position, Command, and Scoped Cell. The table contains one row: 1, create_clock -period 5.000 -name clk -waveform (0.000 2.500) [get_ports clk]. The 'Timing Summary' window is open, showing the 'Design Timing Summary' report. The report is divided into three sections: General Information, Setup, and Hold. The 'General Information' section shows 'Design Timing Summary' with a red warning icon. The 'Setup' section shows 'Worst Negative Slack (WNS): -1,014 ns', 'Total Negative Slack (TNS): -17,970 ns', 'Number of Failing Endpoints: 34', and 'Total Number of Endpoints: 487'. The 'Hold' section shows 'Worst Hold Slack (WHS): 0,140 ns', 'Total Hold Slack (THS): 0,000 ns', 'Number of Failing Endpoints: 0', and 'Total Number of Endpoints: 487'. The 'Pulse Width' section shows 'Worst Pulse Width Slack (WPWS): 2,000 ns', 'Total Pulse Width Negative Slack (TPWS): 0,000 ns', 'Number of Failing Endpoints: 0', and 'Total Number of Endpoints: 392'. The report concludes with 'Timing constraints are not met.'

Position	Clock Name	Period (ns)	Rise At (ns)	Fall At (ns)	Add Clock	Source Objects	Source File
1	clk	5.000	0.000	2.500	☐	[get_ports clk]	mega_add

Position	Command	Scoped Cell
1	create_clock -period 5.000 -name clk -waveform (0.000 2.500) [get_ports clk]	

General Information	Setup	Hold	Pulse Width
Design Timing Summary	Worst Negative Slack (WNS): -1,014 ns	Worst Hold Slack (WHS): 0,140 ns	Worst Pulse Width Slack (WPWS): 2,000 ns
Clock Summary (1)	Total Negative Slack (TNS): -17,970 ns	Total Hold Slack (THS): 0,000 ns	Total Pulse Width Negative Slack (TPWS): 0,000 ns
Methodology Summary	Number of Failing Endpoints: 34	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Check Timing (68)	Total Number of Endpoints: 487	Total Number of Endpoints: 487	Total Number of Endpoints: 392

Timing constraints are not met.